

Technology Stack

Date	29 September 2026
Team ID	6ab3e6756603573ccfb3b891
Project Name	FitBuddy - AI Fitness Plan Generator Using Gemini Models
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML + CSS + Jinja2	Simple server-rendered UI that is easy to maintain and responsive for browser use.
2	Backend / Server-Side	Python + FastAPI	Type-safe request handling, routing, validation and automatic API documentation.
3	Database / Data Storage	SQLite + SQLAlchemy	Lightweight relational storage suitable for the current project and easy local setup.
4	Cloud / Hosting / Deployment	Local Uvicorn deployment; future cloud hosting	Current project runs locally; deployment can later move to a managed platform or container.
5	Version Control & CI/CD	GitHub + Pytest	Repository is available from the existing documentation; automated tests cover basic endpoints.
6	Third-Party APIs / Tools	Google GenAI / Gemini	Generative AI layer produces the workout plan, nutrition tip and feedback-based revision.

Project package dependencies

FastAPI, Uvicorn, Jinja2, SQLAlchemy, python-multipart, python-dotenv, google-genai, Pydantic, Pytest and HTTPX are declared in requirements.txt.